The Arthritis, Autoimmune, and Related Diseases Knowledge (ARK) Portal

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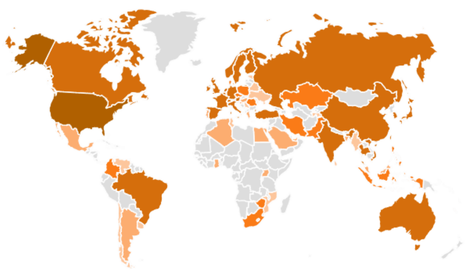
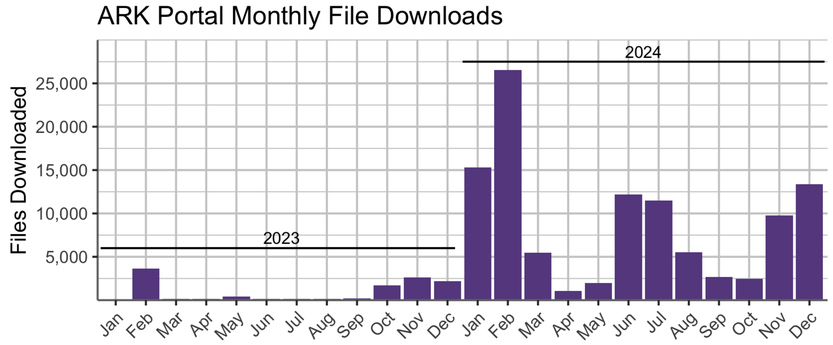
Background

The ARK Portal hosts data generated by a network of research teams working collaboratively to deepen the understanding of Arthritis and Autoimmune and Related Diseases. It was established by the National Institute of Arthritis and Musculoskeletal and Skin Diseases (NIAMS) and includes data from the Accelerating Medicines Partnership® (AMP®)

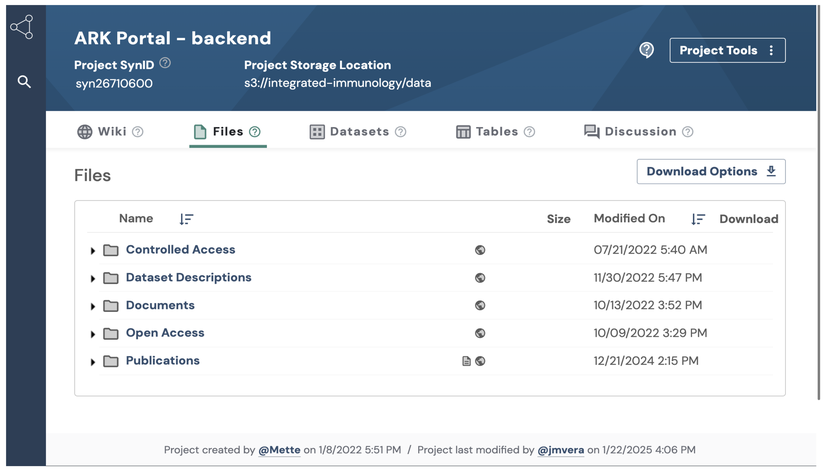


COUNTRY	ACTIVE USERS
United States	2.8K
China	397
United Kingdom	212
Japan	173
India	110
Germany	96
Türkiye	82

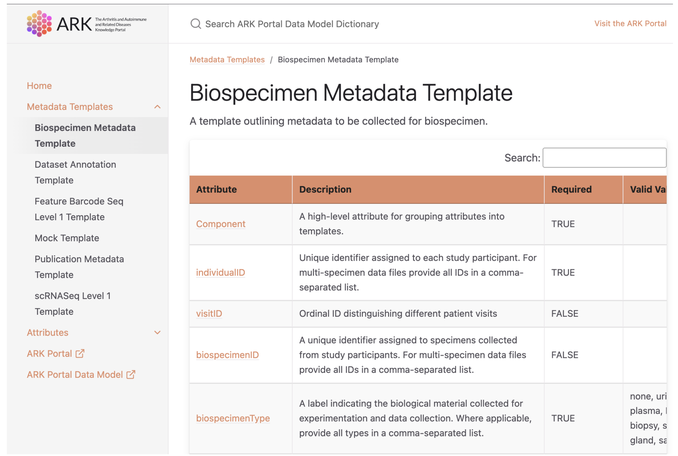
ARK Portal Monthly File Downloads



Backed By Synapse



Metadata Standards

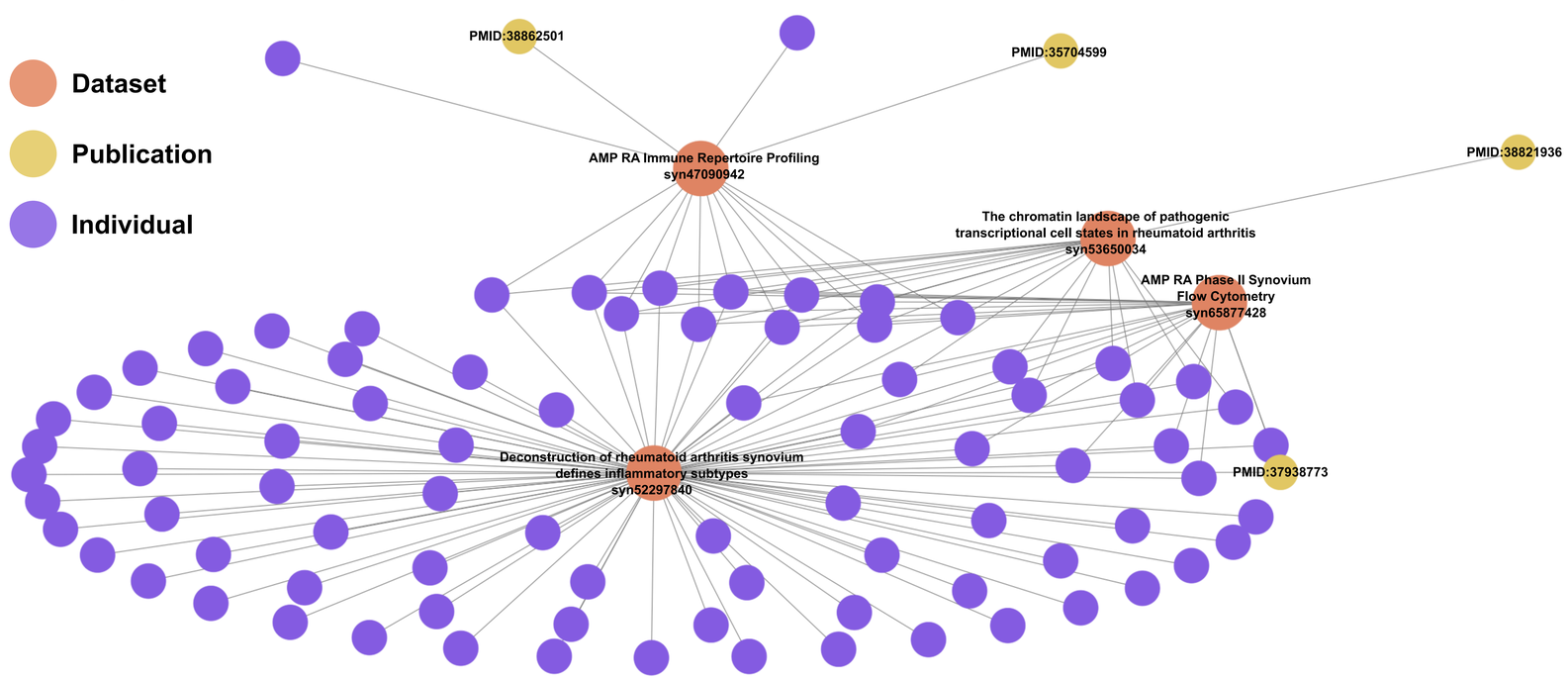


https://ark-portal.github.io/data_model/

Attribute	Description	Valid Values	DependsOn	Properties	Required
Component	A high-level attribute grouping variable.				TRUE
SDP Test Template	A component for use as a mock template for defining file annotation metadata for an SDP.		Component, Filename, fileFormat, resourceType, sampleManifestSynID		
CyTOF SDP Annotation Template	A component for collecting file-level metadata, i.e., annotations, for AMP AIM CyTOF standard data package (SDP).		Component, Filename, fileFormat, resourceType, sampleManifestSynID, targetPanel, targetPanelSynID		
Filename	Name of a file. Must be included in DependsOn for all file manifest/ annotation templates.				TRUE
fileFormat	Standard fileFormat name or file extension	csv, tsv, txt, xlsx, xls, fcs, docx, erate, rec, info, vcf, dose, tbi, fam, bim, bed, bam, h5, mtx, bai, rds, pdf, fgz, zip, fastq, parquet, geojson, tar.gz, zip, omiq			TRUE
Clinical Metadata Template	Clinical data about each study participant		Component, subjectID, visitID		
Biospecimen Metadata Template	Metadata about biospecimens collected from study subjects		Component, subjectID, visitID, projectSampleID, childPSID, slidePSID		
Patient	Patient Component		Component, subjectID		
Biospecimen	A patient biological sample		Component, projectSampleID		
Clinical	Clinical Component		Component, subjectID, visitID		
subjectID	Unique identifier assigned to each study participant. Combination of disease team, site ID, and 3-digit ID				TRUE
visitID	ID associated with specific type of patient visit				TRUE
projectSampleID	Project Sample ID, aka PSID, is a unique biospecimen identifier, the format can vary depending on the specimen or container type.				TRUE
resourceType	High level classification of a file's content	experimental data, metadata, code			TRUE
biospecimenType	Label indicating the tissue of biological source of the sample being profiled.	none, urine, stool, whole blood, serum, plasma, PBMCs, total leukocytes, kidney biopsy, synovial fluid, skin biopsy, salivary gland, saliva, skin swab, synovial tissue			TRUE

ARK Portal Knowledge Graph

Using metadata linkages, e.g., `associatedPublication`, we can build knowledge graphs that connect key resource components from the portal such as publications, datasets, and study subjects (aka individuals). Below is a subgraph depicting the connections between 84 AMP RA/SLE individuals with either rheumatoid arthritis or osteoarthritis who donated synovial tissue that was used in a series of Phase II experimental data collections



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https://arkportal.synapse.org/